He, Yi

Tel: 44-7436418679 E-mail: heylly525@gmail.com Google Scholar Projects

As a machine learning PhD student graduating 2027, I am interested in world models in aspects of efficient multi-modal representation, 3D state reconstruction, learning dynamics and consistent video generation. I am also the main research engineer in our UCL lab. Besides, I am volunteering in software developing a LLM-agent text-to-icon translation app for special education teachers and learners; A *LEGO* enthusiast; A ski-holic over 6 mountains.

EDUCATION BACKGROUND

University College London (UCL)**, Faculty of Engineering, Dynamic Systems Lab** **06/2023 - 2027**

- Program: PhD. AI for Dynamical Systems (Dean's Prize Scholarship awarded)

University College London (UCL)**, Dept. of Computer Science** **09/2020 - 09/2021**

- Program: MSc. Data Science and Machine Learning (Distinction, Dean's List awarded for academic excellence)

University of Glasgow (UoG)**, School of Engineering** **09/2016 - 06/2020**

- Program: BEng (Honors) in Electronics and Electrical Engineering (Academic Excellence scholarship awarded)

PUBLICATION

Conferences

- **He, Y**.*, Zhao, Z.*, Yang, Y., Cheng, X., He, C., Hu Y. (2026). MMPD-Bench: Bridging Multimodal Fission with Multi-Polarimetric Modalities Decomposition. International Conference on Machine Learning. PMLR, 2026.
- Cheng, X.*, Yuan, W.*, Yang, Y., Zhang, Y., Cheng, S., **He, Y**., Sun Z. (2026). Information Shapes Koopman Representation. In The Fourteenth International Conference on Learning Representations, proceedings.
- Li, K., Yang, Y., Cheng, X., He, Y., Sun, Z. (2026). Multilevel Control Functional. In The Fourteenth International Conference on Learning Representations, proceedings.
- **He, Y**.*, Yang, Y.*, Cheng, X.*, Wang H., Xue X., Chen B., Hu Y. (2025). Chaos Meets Attention: Transformers for Large-Scale Dynamical Prediction. International Conference on Machine Learning. PMLR, 2025.
- Yang, Y., Cheng, X., Giles, D., Cheng, S., **He, Y**., Xue, X., Cheng, B., Hu, Y. (2025). Tensor-Var: Efficient Four-Dimensional Variational Data Assimilation. International Conference on Machine Learning. PMLR, 2025.
- Cheng, X.*, **He, Y**.*, Yang, Y., Xue, X., Cheng, S., Giles, D., Tang, X., & Hu, Y. (2024). Learning chaos in a linear way. In The Thirteenth International Conference on Learning Representations, proceedings.
- (Preprint) Yang, Y., Cheng, X., **He, Y**., Li, K., Yuan, W., Sun, Z. (2026). On Stability and Robustness of Diffusion Posterior Sampling for Bayesian Inverse Problems.

Journals

- **He, Y.,** Hu Y. (2025). Constrained factorized dilated temporal convolutional networks for process gases management in steel manufacturing. Advanced Engineering Informatics, 2026.
- Dai, Y., Li, K., Chai, J., Xiao, Z., Yan, B., **He, Y**., & Peng, W. (2021). Radar update: A pedestrian navigation system with enhanced trajectory performance. In 2021 IEEE International Symposium on Circuits and Systems (pp. 1-5).

Patent (20190876262.7): Pedestrian self-adaptive zero-speed updating point selection method based on neural network

ACADEMIC AND RESEARCH EXPERIENCES

Researcher, Dynamics Systems Lab, UCL **11/2024 – Present**

I lead research projects and conducted Dataset and Methodology benchmarking.

- Generated and maintained datasets over 2TB for AI4Science research from multiple physics solvers
- Benchmarked high-resolution Turbulence Channel Flow dataset, Kolmogorov Flow dataset, and Mueller Matrix Polarisation dataset for the AI4Science community
- Published & maintained projects of PFNN, ChaosMeetsAttention and MMPD-Bench.
- Lead 3 papers published to ICML and ICLR, and contributed to other papers in the team.

Research Assistant, Metals Research Institute, Swerim AB 12/2022 – 11/2024

*AI predictive agent in Steel Plant System (***Advisor***: Prof. Chuan Wang, Associate Prof. Yukun Hu)*

- Designed the data pipeline prototype from the gas flow forecasting model to both local storage and cloud storage
- Visualized the real-time system data prediction from Backend model to Front-end data report
- Maintained the project repository and merged the monthly pull requests

Outcome: A full length journal paper accepted in a top journal in information systems (CiteScore **15.8**. IF **9.9**)

MSc Project, Department of Computer Science, UCL AI Center 06/2021 – 09/2021

*Graphical Models and Maximum Entropy in Variational Reinforcement Learning (***Advisor***: Prof. David Barber)*

- Conducted literature review and designed a new Graphical Model and conducted experiments on RL tasks
- Visualized the real-time learning performance with Tensor Board during experiments

Outcome: achieved an outstanding score (78%) for this dissertation

Summer Research, Wireless Sensing Group, James Watt Engineering School, UoG 07/2019 - 09/2019

Signal imaging in human gesture identification with SPFT and Deep Learning classification

INDUSTRY EXPERIENCES

Technology Codecamp, Software Engineer Track, Morgan Stanley UK 08/2022 – 02/2023

- Push and compare updates through Git for agile development, and maintain the development repository on GitHub
- Developed recommendation systems, account management databases, Web Apps and backend programmes

Risk Advisory, RSM UK 01/2022 - 08/2022

- Providing data analytics and risk advisory services for clients from public sectors in the UK.

PRESENTATION & TALKS

Jaguar Land Rover — Warwick, UK

Presentation of AI Technology in Linearizing Nonlinear systems — *07/2025*

University of Oxford – Vectorial Optics and Photonics Group — London, UK

Presentation of Foundation models in AI for Science and Polarization Analysis — *07/2025*

University College London – Department of CEGE — London, UK

3-Minute Paper of Chaos and Nonlinear Dynamics — *04/2025*

University College London – ISI institute — London, UK

Seminar talk of AI for Energy Intensive Digital Twin — *04/2024*

Swerim – The metals research institute — Luleå, Sweden

Presentation of Intelligent Digital Twins for Process Engineering — *10/2023*

The International Conference on Energy, Ecology and Environment 2023 — London, UK

Presentation of AI-assisted energy-rich gas management in manufacturing practices — *08/2023*

SKILLS

- **Deployment skills**: Cluster Computing | Git | ML Cloud Workflow (Google Cloud Platform, Microsoft Azure)
- **Scientific computing**: Numerical Computation with Python (JAX, NumPy, PyTorch)
- **Software designs**: WebApps, SQL Database and Static webpage designs
- **Languages**: Mandarin (Native), English (Professional), Cantonese (Basic), Japanese (Basic)

VOLUNTEERING WORKS

Text-to-icon translator/generator with LLM and RAG for special education needs.

Reviewer of ICML (Gold), NeurIPS and ICLR.

Coordinator in The International Conference on Energy, Ecology and Environment 2023